

Lesson Notes

MODULE 3 | LESSON 1

PORTFOLIO RETURNS AND STANDARD DEVIATIONS

Reading Time	75 minutes
Prior Knowledge	Return, Variance, Standard Deviation, Stocks
Keywords	Mark-to-market, Mark-to-model, Linearity, Non-linearity

In the previous module, we looked at how to compute returns and volatilities of individual securities. We would like to extend this idea to portfolios of securities. We will need to upgrade these calculations to combine two or more returns or two or more variances. In doing so, we will discover the difference between linearity and non-linearity properties. We will see that the key role is played by correlation, which we study in this module. This will lead us to a new security, the exchange-traded fund, which is a security containing other securities.

1. Portfolio Return

In the previous lessons, we looked at how to compute returns. Note that we did this at the individual security level. Now, we turn to portfolios. As you know, a portfolio is simply a collection of securities. For this module, we will imagine combinations of two securities as constituting a portfolio. Indeed, portfolios can contain anywhere from one to thousands of securities. What is the difference between a security and a portfolio? Well, the answer is the number of items. Think of a security as a single asset. Think of a portfolio as a collection of assets. While this may seem simple, you will learn later in this module that a single security could indeed be a collection of other securities. We will cover these exchange-traded funds (ETFs) later in this lesson. Also, we could take a thousand different mortgages and combine them into a single security called a mortgage-backed security (MBS). We will cover these in a future module. An ETF is considered a security, although when you examine its contents, it contains numerous securities.

For this lesson, we will consider a portfolio to be two stocks or two cryptocurrencies. We want to apply the same ideas of return and risk at the portfolio level. Let's create some simple examples.

Suppose the expected return of Stock A is 5%. Suppose the expected return of Stock B is 11%.

What is the expected return of a portfolio that consists of A and B?

First, we might be tempted to take the average. In this case, we see that the expected portfolio return is $(5 + 11)/2 = 8$. Suppose stock A makes up 99% of the

portfolio and stock B makes up 1%. Is the average still appropriate? No. We need to consider the amount each stock contributes to the overall weight. In other words, what proportion is held in stock A? What proportion is held in stock B?

If the portfolio is 60% A and 40% B, then if we let r_P denote the expected portfolio return and w_A, w_B the weights of A and B \$ in the portfolio, respectively, then we have

$$r_P = w_A \times r_A + w_B \times r_B = 0.60 \times 5\% + 0.40 \times 11\% = 7.4\%$$

Notice that the expected return is less than our simple average. Why? We are overweighted on the security that has a lower expected return, so the portfolio's expected return decreases. Suppose the portfolio favors B, with 60% B and 40% A. Then

$$r_P = 0.40 \times 5\% + 0.60 \times 11\% = 8.6\%.$$

This time the expected portfolio return, 8.6 is higher than the simple average of the returns of the two assets A and B.

In the context of portfolios, it is possible for weights to be negative. This is equivalent to selling short an asset. However, we require that the sum of weights is equal to 1. If we assume that $w_A = -0.2$, then $w_B = 1.2$, it's like we shorted asset A and used the proceeds from that short to buy more of asset B. In this case, we would have

$$r_P = -0.2 \times 5\% + 1.2 \times 11\% = 12.20\%.$$

We define this situation as **leveraging the portfolio**. Essentially, we are borrowing money by selling short asset A (the asset with the lower return) and using the proceeds to buy more of asset B (the asset with the higher return). This can be done, but in many situations, a portfolio manager could be prevented from having short positions as clients may consider shorting stocks too risky for their personal risk tolerance. Besides, in our example above, things went well, but we don't usually know what returns we can achieve, and we could actually lose money on asset B, in which case the use of leverage would magnify our negative return.

It's also possible for the weights of a portfolio to sum to zero. In this case, it would be like assuming that we built a portfolio where we are short an asset and we are long the other asset, and the two dollar amounts are the same. We call this portfolio a **dollar-neutral** portfolio. This would be akin to investing without using one's money. Some hedge funds may come close to having positions like this, but for our course now, this is not something to consider.

In a portfolio, weights represent the proportion of market value and not merely the number of shares that we invested in the assets. So let's write the formula in terms of number of shares and prices:

$$r_P = \frac{q_A \times p_A \times r_A + q_B \times p_B \times r_B}{q_A \times p_A + q_B \times p_B}$$

- q_A = number of shares of stock A
- p_A = share price of stock A
- r_A = expected return of stock A
- q_B = number of shares of stock B

- p_B = share price of stock B
- r_B = expected return of stock B

In other words, the weights represent a proportion of market values. A further way to write the formula for the portfolio return is

$$r_P = \frac{MV_A \times r_A + MV_B \times r_B}{MV_A + MV_B}.$$

2. Portfolio Standard Deviation

In this section, we are going to look at portfolio variance. The formula for portfolio variance is not as simple as the one for portfolio return. In fact, if we indicate the portfolio variance with the symbol σ_P^2 , then

$$\sigma_P^2 = w_A^2 \times \sigma_A^2 + w_B^2 \times \sigma_B^2 + 2\rho_{A,B}w_A \times w_B \times \sigma_A \times \sigma_B.$$

You will recognize the terms representing weights. There are two new terms:

- σ_A represents the standard deviation (volatility) of stock A;
- σ_A^2 represents the variance of stock A;
- The corresponding terms apply to stock B;
- $\rho_{A,B}$ is a new term that represents the **correlation** between the returns of stock A and stock B.

Know that there are three terms on the right-hand side. Let's examine each term individually:

- The first term on the right-hand side represents the variance of stock A, but note that its coefficient is not merely the weight but rather the weight squared. This square exists because we are in the realm of variance, not standard deviation.
- The second term represents the square-weighted variance coming from stock B. The key point about these terms is that they are always positive. Any number squared here is positive (there are no imaginary weights!). The sum of positives will be positive.
- The third term on the right-hand side represents the interaction of the two stocks. This term itself consists of five inputs: the correlation coefficients, two weights, and two standard deviations.

NOTE. Let's reiterate that we are using long-only portfolios, so all the weights, along with all the standard deviations, are positive.

The correlation value is the only number on the entire right-hand side that does not have to be positive (assuming weights are positive rather than 0 to indicate no position in that stock). It turns out that the correlation plays a key role in determining the portfolio's volatility. We'll have much more to say about this, but for now, let's get some numbers. Since it's helpful to think in terms of standard deviation, let's take the square root of both sides and derive the formula for a portfolio's standard deviation:

$$\sigma_P = \sqrt{w_A^2 \times \sigma_A^2 + w_B^2 \times \sigma_B^2 + 2\rho_{A,B}w_A \times w_B \times \sigma_A \times \sigma_B}.$$

Example. Suppose we have an equally weighted portfolio between A and B and that $r_A = 5\%$, $r_B = 11\%$, $\sigma_A = 3\%$, $\sigma_B = 8\%$, and $\rho_{A,B} = 0.7$. What is the portfolio standard deviation?

Answer. If we use the formula just introduced above, we find

$$\sigma_P = \sqrt{0.50^2 \times 0.03^2 + 0.50^2 \times 0.08^2 + 2 \times 0.70 \times 0.50 \times 0.50 \times 0.03 \times 0.08} = 5.162\%.$$

The portfolio's standard deviation appears to be like the expected return, in that the value falls between the range of the individual securities. However, this would be an incorrect interpretation. We'll revisit this topic later in this lesson. For now, let's see how we would combine the expected return and standard deviation.

3. Portfolio Return and Variance

Recall from Module 2 that we took a stock's volatility and divided it by the stock's expected return. This is the coefficient of variation. We can do the same for a portfolio. Using CV_P to denote the coefficient of variation for the portfolio, we have

$$CV_P = \frac{\sigma_P}{r_P}.$$

In our previous example, we have a coefficient of variation of

$$CV_P = \frac{5.162\%}{8\%} = 0.645.$$

We can use this number to compare it to other portfolios and determine the number of volatility points we pay per unit of return. As always, the smaller this number, the greater the "*bang for the buck*." Generally, this number is used in relative comparisons rather than absolute comparisons. Can we compare it to Security B itself? For Security B,

$$CV_B = \frac{8\%}{11\%} = 0.727$$

Interestingly, the portfolio with A and B has a lower "vol per unit of return cost" than stock B alone. By including stock A in the portfolio, the portfolio somehow has a better risk-return relationship than B alone.

4. Portfolio Sharpe Ratio

The **Sharpe ratio** for a portfolio, which we call SR_P , is also easily calculated:

$$SR_P = \frac{r_P - r_{rf}}{\sigma_P},$$

where

- r_P is the portfolio return;
- r_{rf} is the risk-free rate; and
- σ_P is the portfolio volatility.

If we assume that $r_{rf} = 1\%$, then the Sharpe ratio for the portfolio of our example is found to be

$$SP_P = \frac{8\% - 1\%}{5.162\%} = 1.356$$

Let's compare it to the Sharpe ratio of stock A and B this time:

$$SR_A = \frac{5\% - 1\%}{3\%} = 1.333$$

and

$$SR_B = \frac{11\% - 1\%}{8\%} = 1.25$$

Interestingly, the portfolio with A and B has a higher Sharpe ratio, that is, a higher return per unit of risk/volatility than stock A alone. By including stock B in the portfolio, the portfolio seems to have a better risk-return profile than stock A alone. The same holds for stock B.

5. Portfolio Return and Variance

The calculations we did are in no way specific to stocks. Indeed, this method can be applied to any asset class. Let's discuss some important issues regarding the weights.

The weights represent proportions of market value. The portfolio returns and variance can be used even if the asset classes are different. The weights should reflect the appropriate market value at a given time. Let's examine the formula for weights.

$$w_A = \frac{MV_A}{MV_A + MV_B} = \frac{q_A \times p_A}{q_A \times p_A + q_B \times p_B}$$

When A and B are two stocks, it is likely very easy to ensure that they have observable prices. But what if we combined one asset that is exchange traded and another that rarely trades, such as real estate? Since exchanges have official closes, it is easy to get the stock price correct. Each day we can mark our position using the market's close. This process is known as **mark-to-market**. We mark our security prices to levels indicated by the market. But how would we know the price of an asset without exchange closes or OTC closing prices? We may need a **mark-to-model** price. We would mark our security at levels indicated by a model, provided by our company or perhaps even regulatory requirements. We would need a model to indicate the price, perhaps as often as daily, so we can compute not only expected returns but also volatilities. Without a consistent price series, it is not possible to compute returns, let alone volatilities.

Once we're sure that we have reliable levels, we can then compute returns and volatilities and combine them into their portfolio equivalents. One other consideration reflects that we live in a global market, with different exchanges opening and closing at different times. To be precise, we should find a time at which we agree to compute returns—for example, at the close of business London time. This is known as Greenwich Mean Time (GMT) and is often used as a reference point for time.

Now that we've discussed the weights, let's revisit the properties of return and variance.

6. Portfolio Return and Standard Deviation (Revisited)

One key property of portfolio return is that it is a linear form of the returns of assets.

In particular, if we have two assets, assume that weights cannot be negative and the weights sum to 1, then:

- The portfolio return is always going to be in between the returns of the two assets.
- If we set one of the weights to zero, then the portfolio return will equal the return of the other asset.

If we allow for a negative weight, then the return on the portfolio can exceed the return on the other asset, which we saw in an earlier example. However, the linearity property still holds.

The portfolio standard deviation is non-linear. By looking at the formula, we see two complications that invalidate linearity. Indeed, portfolio variance is a non-linear function of the individual variances:

- The summation is under a square root sign.
- There is a third term that comes into play whenever correlation is non-zero.

When two assets are negatively correlated, they tend to move in opposite directions, i.e., if one asset grows in value, the other tends to decrease in value and vice versa. This should make the value of the portfolio less volatile.

Example. Consider the same portfolio that we've used several times lately and assume that the portfolio is equally weighted between asset A and asset B. However, this time, we assume that the returns of A and B have a negative correlation. More specifically, let $\rho_{A,B} = -0.8$.

Using the formula, we find that

$$\sigma_P = \sqrt{0.50^2 \times 0.03^2 + 0.50^2 \times 0.08^2 + 2 \times (-0.80) \times 0.50 \times 0.50 \times 0.03 \times 0.08} = 2.94\%.$$

In this case, the volatility of the portfolio is lower than that of both asset A and asset B.

This last example tells us that if we can find two securities that are strongly negatively correlated, we would say that one *hedges* the other. That is, by using A and B in this case, we are reducing the overall risk (as well as the return because it's typically not possible to decrease risk without accepting a reduction in return as well). Indeed, the negative correlation is the only way that the variance of a long-only portfolio can be less than the variance of each individual security's variance.

From a pedagogical point of view, we have limited ourselves to considering a portfolio consisting of just two assets because this makes it easier to illustrate the main features that matter to us. However, we could easily consider a portfolio that consists of N assets. In this case, the formula for portfolio return remains more or less equally simple, i.e.:

$$r_P = w_1 r_1 + w_2 r_2 + \dots + w_N r_N = \sum_{i=1}^N w_i r_i.$$

The formula for the standard deviation of the portfolio is a little bit more complex but not dramatically so:

$$\sigma_P = \sqrt{\sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_i \sigma_j \rho_{i,j}}.$$

The general formulae with $N \geq 2$ assets will be needed in the Portfolio Management course, among other places. In fact, in that course, we will see how to express returns and variances/standard deviation in terms of vectors and matrices.

7. Conclusion

In this lesson, we examined the returns and standard deviations of portfolios. There seems to be some kind of magic involved when using portfolios compared to individual stocks. Adding more stocks seems to get good returns but without as high of a risk. The key player will be correlation. In the next lesson, we will closely examine the role correlation plays in shaping those returns.

